Project Title

BlinkIT Grocery Sales Analysis

Domain: Retail & Grocery Sales Analytics

Company: Sapphire IQ

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Ram is working as a Data Analyst at Sapphire IQ and has been assigned a BlinkIT Grocery Sales Analysis project in the Retail & Grocery Sales Analytics domain. The purpose of this project is to study the available grocery sales data and provide management with a clear understanding of business performance across different products, categories, outlets, and locations.

BlinkIT offers a wide range of grocery and daily-use products through different types of outlets. With a large amount of sales information available, management wants to identify which products are performing well, which categories are generating higher sales, which outlets are contributing more to overall sales, and where there is scope for improvement.

To address these requirements, Ram will use Python to organize, clean, explore, and analyze the available sales data. The analysis will focus on identifying important patterns, trends, and differences in sales performance. Factors such as product type, product characteristics, outlet type, outlet size, outlet location, customer ratings, and product demand will be examined to understand their relationship with overall sales.

Ram will compare the performance of different products, categories, and outlets to identify high-performing and low-performing areas. The results will then be presented through meaningful summaries and visualizations so that management can easily understand the key findings.

The final analysis will help Sapphire IQ's management team make better decisions related to product selection, outlet performance, inventory planning, sales improvement, and future business strategies. The project will transform raw grocery sales data into clear and actionable business insights that can be easily understood by both technical and non-technical stakeholders.

Objectives and Tasks

Objective 1: Understand the Business and Its Sales Performance

Task 1.1: Understand what types of products BlinkIT is selling.

Task 1.2: Understand the different types and locations of BlinkIT outlets.

Task 1.3: Understand how much BlinkIT is selling overall.

Task 1.4: Understand the overall customer rating and product performance.

Objective 2: Make Sure the Available Information Is Reliable

Task 2.1: Find information that is missing from the available records.

Task 2.2: Find information that has been repeated more than once.

Task 2.3: Find values that look unusually high or unusually low.

Task 2.4: Make sure similar information is presented in the same way for proper comparison.

Objective 3: Find Out What Is Driving Better Sales Performance

Task 3.1: Find out which types of products are selling more.

Task 3.2: Find out which types of outlets are performing better.

Task 3.3: Understand whether product visibility, weight, ratings, and outlet characteristics are connected with sales.

Task 3.4: Compare better-performing and lower-performing products and outlets to understand the difference.

Objective 4: Help Management Make Better Business Decisions

Task 4.1: Identify the products that are contributing more to sales.

Task 4.2: Identify the outlets and locations that are performing better.

Task 4.3: Identify products, outlets, and areas where performance can be improved.

Task 4.4: Provide practical suggestions to improve sales and overall business performance.